HABTAMU MENBERU TEDILA

Postdoctoral Researcher in Radio Astronomy
McGill University • CHAMPSS • CHIME/Pulsar • CHIME/FRB
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RESEARCH PROFILE

Radio astronomer specializing in pulsar astrophysics, time-domain radio astronomy, and radio-wave propagation through the ionized interstellar medium. At McGill University, I contribute to the CHIME All-Sky Multiday Pulsar Stacking Search (CHAMPSS), where I am developing the CHAMPSS Pulsar Morphology and Propagation Pipeline — an automated framework for characterizing pulse-profile morphology and measuring scattering timescales, dispersion-measure variations, and flux-density variability across daily CHIME observations.

My broader research uses observations from FAST, Parkes, CHIME, and the Nanshan radio telescope to investigate pulsar timing, polarization, single-pulse variability, nulling, mode changing, subpulse drifting, microstructure, scattering, and scintillation. I also contribute to the CHIME/Pulsar and CHIME/FRB collaborations, including studies of the long-term dispersion-measure evolution of repeating fast radio bursts. Before joining McGill, I was a postdoctoral researcher at the National Astronomical Observatories of the Chinese Academy of Sciences.

Research Interests

- **Pulsars:** timing and emission physics; pulse-profile morphology; single-pulse analysis; polarization
- **Propagation:** interstellar scattering and scintillation; dispersion-measure and flux-density variability
- **Transients & GW:** fast radio bursts (FRBs); gravitational-wave detection with pulsars
- **Methods:** scientific pipeline development for large radio surveys

ACADEMIC APPOINTMENTS

Postdoctoral Researcher in Radio Astronomy Feb. 2026 – present
McGill University, Montréal, Canada
Trottier Space Institute Postdoctoral Fellow
Supervisor: Prof. Victoria Kaspi

Postdoctoral Researcher in Radio Astronomy May 2023 – Jan. 2026
National Astronomical Observatories, Chinese Academy of Sciences (NAOC), Beijing, China
Supervisor: Prof. Di Li

Lecturer in Physics Sep. 2014 – Aug. 2019
Arba Minch University, Arba Minch, Ethiopia

Graduate Assistant I & II in Physics Sep. 2011 – Aug. 2014
Arba Minch University, Arba Minch, Ethiopia

EDUCATION

Ph.D. in Astrophysics Aug. 2019 – January 2023
University of Chinese Academy of Sciences, Beijing, China
Thesis: Extreme switching and emission variation in radio pulsars
Supervisor: Prof. Na Wang

M.Sc. in Astrophysics Sep. 2012 – Oct. 2014
Addis Ababa University, Addis Ababa, Ethiopia
Thesis: Synchrotron radiation from Crab pulsars

Advisor: Dr. Legesse Wetro Kebede

B.Sc. in Physics Sep. 2008 – Jun. 2011
Debre Markos University, Debre Markos, Ethiopia

FELLOWSHIPS AND AWARDS

2026 Trottier Space Institute Postdoctoral Fellowship
2024 ANSO Early-Career Scientist Visiting Fellowship on Mega-science Facilities
2023 NAOC Postdoctoral Position and Fellowship
2022 Excellent International Student
2019 CAS-TWAS President's Fellowship

PUBLICATIONS

*21 peer-reviewed papers (9 as first or corresponding author) and 6 manuscripts under review. * denotes corresponding author.*

First- and Corresponding-Author Papers (Peer Reviewed)

1. **Tedila, H. M.**, Li, D., Wang, P., Yuen, R., Wu, Z. W., Dang, S. J., Yuan, J. P., Wang, N., et al. Multifaceted Emission Properties of PSR J2129+4119 Observed with FAST. *ApJ, 997(1), 24 (2026).*
2. **Tedila, H. M.**, Li, D., Wang, P., Yuen, R., Yuan, J. P., Wang, N., et al. Unveiling the emission properties of three long-period pulsars using FAST. *ApJS, 277(2), 39 (2025).*
3. Cai, Y., Dang, S. J., **Tedila, H. M.***, Hao, Y., Li, W., Yuan, J. P., et al. A single-pulse study of PSR J0535−0231 discovered by FAST. *MNRAS, 544(1), 299–308 (2025).*
4. **Tedila, H. M.**, Yuen, R., Wang, N., Li, D., et al. Investigation of profile shifting and subpulse movement in PSR J0344−0901 with FAST. *ApJ, 965(2), 144 (2024).*
5. Li, Q., Dang, S., Shang, L., **Tedila, H. M.***, et al. Subpulse drifting of PSR J1514−4834. *Ap&SS, 369, 90 (2024).*
6. **Tedila, H. M.**, Yuen, R., Wang, N., et al. Emission variation of a long-period pulsar discovered by the Five-hundred-meter Aperture Spherical Radio Telescope (FAST). *ApJ, 929(2), 171 (2022).*
7. **Tedila, H. M.** Magnetic dipole interaction with multipole magnetic field lines of neutron stars. *Indian J. Phys., 96(3), 689–695 (2022).*
8. **Tedila, H. M.**, Yuen, R., Han, X. H. Properties of the emission region in pulsars with opposite subpulse drift directions in different profile components. *Ap&SS, 366(10), 100 (2021).*
9. **Tedila, H. M.** Multipole magnetic field lines interaction of neutron stars. *Research & Reviews: Journal of Physics, 9(2) (2020).*

Co-Authored Papers (Peer Reviewed)

1. Cui, X. H., Zhang, C. M., Li, D., Wang, P., et al. (incl. **Tedila, H. M.**) Indication for Decreasing Dispersion Measure in the Population of Repeating Fast Radio Bursts and Connection to Young Supernova Remnant Expansion. *ApJ, 1007(2), 173 (2026).*
2. Khasanov, S., Safarov, A., **Tedila, H. M.** Multi-probe tests of Lorentz-symmetry-violating gravity from black hole observations to cosmological tensions. *Physics of the Dark Universe, 102353 (2026).*
3. Wu, Q. D., Yuan, J. P., Wang, N., Li, D., Wang, P., et al. (incl. **Tedila, H. M.**) Correction to: Follow-up timing of 24 pulsars discovered in Commensal Radio Astronomy FAST Survey. *MNRAS, 545(1), staf1986 (2026).*
4. Yuen, R., **Tedila, H. M.** Changes in emission properties associated with the emission shifts in PSR J0344−0901. *Universe, 11(5), 165 (2025).*

5. Rejep, R., Zhu, W. W., Wang, N., et al. (incl. **Tedila, H. M.**) A detailed single-pulse study of four-component pulsar PSR J1239+0326 discovered by FAST. *ApJ, 985(1), 11 (2025).*
6. Khasanov, S., Su, Y., Safarov, A., **Tedila, H. M.**, Mamatkulov, O. Major radioactive ion beam facilities: A brief global overview. *Eur. Phys. J. Plus, 140(7), 638 (2025).*
7. Dang, S. J., Yuan, J. P., Shang, L. H., Xu, X., **Tedila, H. M.**, et al. Polarization and single-pulse microstructure studies of Rotating Radio Transient J0139+3336 with FAST. *MNRAS, 528(2), 1213–1222 (2024).*
8. Cai, Y., et al. (incl. **Tedila, H. M.**) The study of mode switching behavior of PSR J0614+2229 using the Parkes ultra-wideband receiver observations. *ApJ, 966(2), 241 (2024).*
9. Wu, Q. D., Yuan, J. P., Wang, N., Li, D., Wang, P., et al. (incl. **Tedila, H. M.**) Follow-up timing of 24 pulsars discovered in the Commensal Radio Astronomy FAST Survey. *MNRAS, 522(4), 5152–5164 (2023).*
10. Miao, C.-C., et al. (incl. **Tedila, H. M.**) Reciprocating magnetic fields in the pulsar wind observed from the Black Widow pulsar J1720−0534. *RAA, 23(10), 105005 (2023).*
11. Wen, Z. G., et al. (incl. **Tedila, H. M.**) A single-pulse study of the subpulse drifter PSR J1631+1252 discovered at FAST. *ApJ, 929(1), 71 (2022).*
12. Khasanov, S., Suleymanov, R., Safarov, A., Safarov, A., **Tedila, H. M.**, Muratov, R. Calculation of exposure and absorbed dose at irradiation samples by electron bremsstrahlung. *Radiat. Phys. Chem., 188, 109651 (2021).*

Preprints and Manuscripts Under Review/Accepted

1. Zhang, J. S., Wang, T. C., Wang, P., Wu, Q., Li, D., et al. (incl. **Tedila, H. M.**) The magnetar model's energy crisis for a prolific repeating fast radio burst source. *arXiv:2507.14707; Nature Astronomy, accepted (2026).*
2. **Tedila, H. M.**, Dang, S., Li, D., Wang, P., Yuan, J. P., Yuen, R., Wang, N., Khasanov, S. Timing, Polarization, and Single-Pulse Properties of Long-Period FAST Pulsars. *arXiv:2608.22771 (2026).*
3. Khasanov, S., Safarov, A., Yang, B., **Tedila, H. M.** Ruppeiner thermodynamic geometry, microstructure, quasinormal modes and greybody factors of the Einstein–Skyrme black hole. *arXiv:2607.05673 (2026).*
4. Jain, N., Shin, K., Kaspi, V. M., et al. (incl. **Tedila, H. M.**) The FRB Population Revisited: Rate, Energy, Evolution, and Host-DM Constraints from the Second CHIME/FRB Catalog. *ApJ, under review (2026).*
5. Bai, J. T., Zhao, D., et al. (incl. **Tedila, H. M.**) Probing the L-band Radio Eclipse of the Black Widow Pulsar PSR J1810+1744 with FAST. *ApJ, under review (2026).*
6. Dang, S., Lu, J., Jiang, P., Liu, Y., Xie, J., **Tedila, H. M.**, et al. A Rare Millisecond Pulsar with Cross-Pole Emission: Single-Pulse Insights from PSR J1857+0943. *arXiv:2512.21892 (2025),* ApJ*, accepted 2026*

AWARDED OBSERVING PROPOSALS

18 awarded proposals on FAST and Parkes: 6 as PI and 12 as Co-PI.

As Principal Investigator

1. Follow-up timing studies for four millisecond pulsars discovered by FAST. PT2025_0207 (2025)
2. Follow-up timing studies for 15 slow-period pulsars discovered by FAST. PT2024_0220 (2024)
3. Temporal and spectral analysis of microstructures in slow-period CRAFTS pulsars. PT2024_0036 (2024)
4. Investigation of bi-drifting subpulses in PSR J0535−0231 using FAST. PT2023_0094 (2023)
5. Investigation of the physics of pulsars with complex emission features. PT2022_0193 (2022)
6. Investigating unusual profile mode changing in new FAST-discovered radio pulsars. PT2020_0052 (2020)

As Co-Principal Investigator

1. Timing of 2 normal pulsars discovered by CRAFTS (PI: F. Liu). PT2026_0225 (2026)

2. Single-pulse study of newly discovered CRAFTS pulsars PSRs J1813−0452 and J0344−0901 (PI: R. Rejep). PT2024_0196 (2024)
3. Long-term timing of 6 CRAFTS pulsars discovered by FAST (PI: R. Rejep). PT2024_0248 (2024)
4. Single-pulse study of two newly discovered CRAFTS pulsars PSR J2159+0202 and PSR J1239+0326 (PI: R. Rejep). PT2023_0179 (2023)
5. Long-term timing of 9 CRAFTS pulsars discovered by FAST (PI: R. Rejep). PT2023_0181 (2023)
6. Polarization and single-pulse microstructure studies of rotating radio transients with FAST (PI: S. J. Dang). PT2023_0129 (2023)
7. Timing of 6 millisecond pulsars discovered by CRAFTS (PI: C.-C. Miao). PT2023_0034 (2023)
8. FAST newly discovered pulsar follow-up timing observation at Parkes telescope (PI: D. Li). PX500 (2023)
9. Timing analysis and emission variations in new pulsars discovered in GPPS (PI: J. P. Yuan). PT2020_0071 (2020)
10. Extreme switching and radio emission variation in FAST pulsars (PI: R. Yuen). PT2020_0045 (2020)
11. FAST observation of four 'Swooshes' pulsars (PI: S. J. Dang). PT2020_0062 (2020)
12. Pulsar timing: Physics and evolution of pulsars (PI: N. Wang). ZD2020_6 (2020)

OBSERVING EXPERIENCE

CHIME — Canadian Hydrogen Intensity Mapping Experiment Feb. 2026 – present
CHAMPSS, CHIME/Pulsar, and CHIME/FRB research

FAST — Five-hundred-meter Aperture Spherical radio Telescope Apr. 2020 – Jan. 2026
Pulsar timing, single-pulse studies, and pulsar emission-mechanism research

Parkes 64-m Radio Telescope (Murriyang) Jun. 2023 – Jan. 2026
Observing expert for the FAST pulsar-timing follow-up project PX500

Nanshan 26-m Radio Telescope Jan. 2020 – Dec. 2022
Pulsar observations and monitoring

SELECTED PRESENTATIONS

2026 AstroQuebec Seminar, Université of Montréal, — *Canada*
2026 Special invited talk, 1st Annual Meeting of the Astronomical Society of Ethiopia
2026 Oral presentation, African Astronomical Society Conference — *Kasane, Botswana*
2025 Oral presentation, 11th Annual Science at Low Frequencies Conference — *Orléans, France*
2025 Oral presentation, Second FAST Scientific Forum — *Guiyang, China*
2024 Oral presentation, International Pulsar Symposium — *Guiyang, China*
2024 Oral presentation, SKA Pulsar Science Symposium — *Kunming, China*
2024 Invited talk, Observational Radio Astronomy and Its Future Development in Ethiopia — *Addis Ababa, Ethiopia*
2024 Poster, Green Bank Observatory Single Dish Summer School — *West Virginia, USA*
2023 Oral presentation, Scientific Program of FAST / Future Pulsar Symposium 12 — *Nanyang, China*
2023 Oral presentation, ISM Science Forum — *Wuhan, China*
2022 Oral presentation, FAST Pulsar Symposium 11 — *Xiangtan, China*

TEACHING AND MENTORSHIP

Courses Taught — Arba Minch University (2011–2019)

- **Introductory:** Mechanics (Phys1011); Electromagnetism (Phys1013)

- **Mathematical Methods:** Mathematical Methods of Physics I & II (Phys2031, Phys2032)
- **Core:** Classical Mechanics I & II (Phys2041, Phys2042); Electrodynamics I & II (Phys3081, Phys3082)

Undergraduate Thesis Supervision — Arba Minch University (11 Undergraduate students)

- **2014–2016:** Dirshaye Fikadu Berga; Yenenesh Zerefu Degefa
- **2015–2017:** Meggarsa Tase Amensisa; Urisa Urage Elemo; Mebrat Senshaw Eskezia
- **2016–2018:** Kelem Tedla Legesse; Seada Hussen Yimam; Hashim Hussen Yesuf
- **2017–2019:** Dereje Chala Abdi; Eyob Endale Kie; Adem Seid Muhie

PROFESSIONAL ACTIVITIES

Journal Refereeing

- Reviewer for *Monthly Notices of the Royal Astronomical Society* (MNRAS)
- Reviewer for *Physics of the Dark Universe*
- Reviewer for *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* (NIM B)

Collaborations

- CHAMPSS; CHIME/Pulsar; CHIME/FRB

Professional Memberships

2025 – Astronomical Society of Ethiopia — *Full member (since Jan. 2025)*

2024 – African Astronomical Society — *Full member (since Jun. 2024)*

2014 – Ethiopian Physical Society — *Regular member (since Sep. 2014)*

REFERENCES

Prof. Victoria Kaspi McGill University; Director, Trottier Space Institute, victoria.kaspi@mcgill.ca

Prof. Di Li Professor and Chief Scientist, National Astronomical Observatories, CAS, dili@tsinghua.edu.cn

Prof. Pei Wang Professor, National Astronomical Observatories, CAS, wangpei@nao.cas.cn

Prof. Na Wang Professor and Director, Xinjiang Astronomical Observatory, CAS, na.wang@xao.ac.cn

Prof. Rai Yuen Professor, Xinjiang Astronomical Observatory, CAS, ryuen@xao.ac.cn